What Drives Movie Success?

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How the project evolved

We started by considering three topics: **Spotify listening data**, **vibe-coding** (how non-technical people are building software with AI), and **movies**. Jiahui pushed for movies — the topic that everyone has an opinion about, with a dataset that lets us look at commercial outcomes, critical reception, and cultural language all at once. The rest of us came around because each of the required visualization types attached naturally: geography (production countries), text (reviews), and networks (cast).

Our first framing was generic: *"What makes a movie successful?"* As we explored the data, we realized the question was hiding an assumption — that "success" was one thing. Box office and audience ratings rarely move together, and our central thesis shifted to:

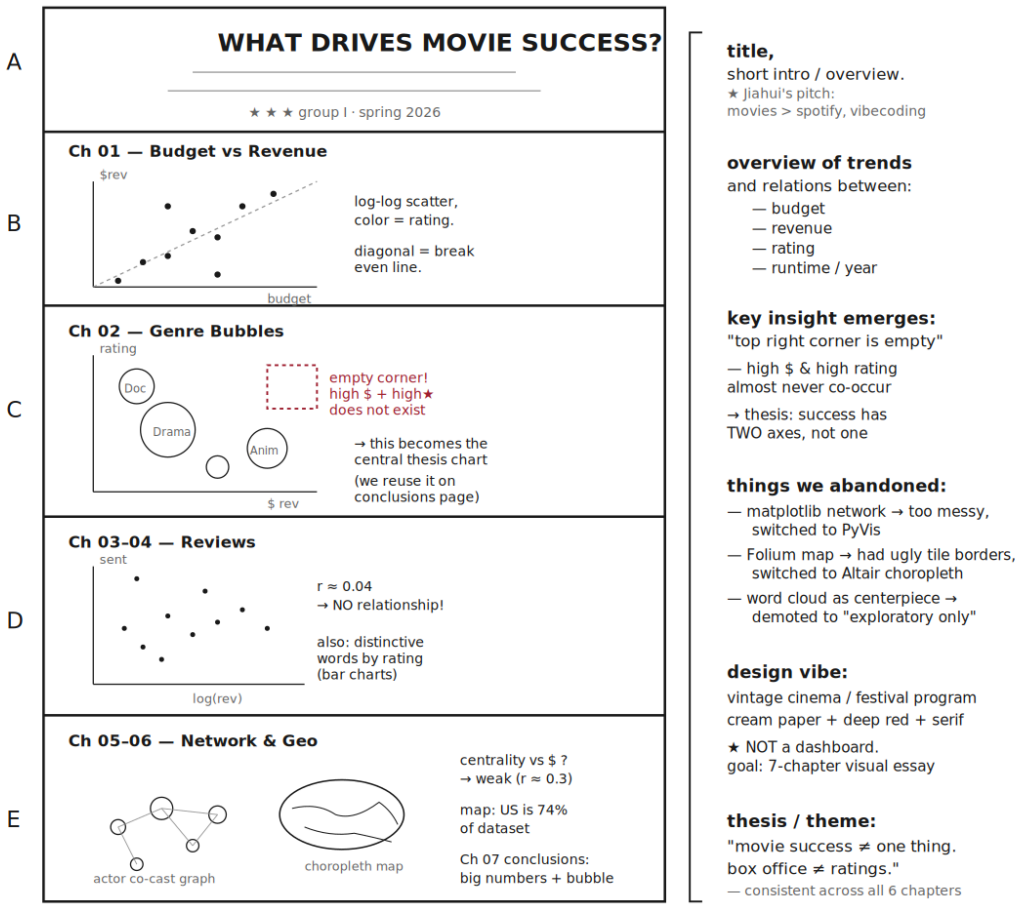
> *"Movie success is not a single dimension. The factors that predict box office are largely different from the factors that predict ratings."*

Tools we ended up using

Streamlit for the multi-page app structure. **Altair** as the primary chart library — chosen because the course covered it extensively and its declarative grammar made interactivity easy. **PyVis** for the actor network (matplotlib produced an unreadable hairball). **Altair choropleth** for the world map (replacing an earlier Folium version that had ugly tile borders). **WordCloud** only as exploratory aid — kept small.

Initial design sketch

Below is our planning sketch: a structural overview of the seven chapters, with notes on the right margin recording the design decisions and abandoned approaches as they came up.



Hand-style sketch of the website's seven-chapter structure, with margin notes tracking what we kept, what we abandoned, and the central thesis as it emerged.

What we abandoned along the way

- **Networkx + matplotlib for the actor network.** Produced a static, unreadable hairball. PyVis's physics simulation made the structure legible and added interaction (drag, zoom, hover).
- **Folium for the world map.** It worked but required a tile-server URL, rendered with branding marks, and clashed with our cinema theme. We rewrote it as an Altair choropleth using world-atlas TopoJSON.
- **Word clouds as a centerpiece.** Visually striking but quantitatively weak. We demoted them to an exploratory aid only and made the comparative bar charts the main analytical evidence in the text chapter.
- **Dashboard layout.** Our app at first looked like most Streamlit projects — sidebar full of filters, charts stacked one after another, no narrative. We restructured it as a seven-chapter visual essay with chapter covers and per-chapter takeaways.

What we'd do differently

- **Sample more reviews per movie.** Many films have only one or two reviews; the sentiment chapter would be more rigorous with a higher minimum threshold. TMDB API rate limits made larger collections slow.
- **Use a transformer-based sentiment model.** VADER is a reasonable baseline but performs modestly on long, mixed-tone reviews ($r \approx 0.4$ vs user ratings). A fine-tuned model could potentially push that higher.
- **Add temporal sentiment.** Right now Chapter 3 treats every review as if written at the same time. A timeline view could reveal which films aged into reappraisal.
- **Director and writer networks.** Our network analysis stops at actors. Director-actor and writer-director collaborations might explain critical success in ways the cast network does not.

What we learned

The biggest single lesson was that **chart choice and narrative choice are not separable**. The genre bubble chart in Chapter 2 isn't powerful because of its encoding; it's powerful because of an empty corner that the surrounding prose tells you to look at. Strip away the narrative and the same chart becomes "a scatter plot of genres."

We also learned that **constraint produces more cohesion than abundance**. One chart library, one color palette, one font pair. The chapters that came together fastest were the ones where we resisted introducing a new chart type.